# Syed Maroof Hussain

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## Experience

**Monero – Secure, Private, Untraceable Cryptocurrency**

Open Source Contributor

- Contributed to the **C++** core of Monero, a 12-year-old **open-source**, **distributed cryptocurrency** with **11,000+ GitHub stars**, **3,500+ forks**, and **27,000+ daily transactions**.
- Implemented **per-peer bandwidth rate limiting**, so no single peer can exhaust a node's bandwidth on low-end hardware; addresses an **8+ year-old** feature request.
- Received **strong positive feedback from core Monero maintainers**; network data showed **62%** of peers use more than 50 kB/s, so the feature directly helps users on **slow internet or lower-end hardware**.
- Fixed a **security vulnerability** where the code ignored **DNSSEC** validation for **6+ years**, closing a **DNS** poisoning attack vector before it could be exploited.
- Wrote **GoogleTest** coverage and **documentation** for every new feature; iterated with core contributors in **code review**, with all changes **approved for the next major release**.

**PCapPlusPlus – C++ Packet Parsing & Creation Library**

Open Source Contributor

- Implemented **packet parsing, creation, and serialization** for the **Bluetooth HCI** protocol in a 12-year-old **open-source C++ network** library with **3,000+ stars**, closing a **3-year-old** feature request.
- Designed the protocol layer using **object-oriented design**, reviewing it **directly with the project owner** and lead maintainers as part of a larger effort to add support for new protocols.
- Contributed multiple PRs alongside dozens of contributors to modernize the codebase to **C++14**, preventing memory leaks to make it more **stable** and **maintainable**.

## Projects

**Price Matching Engine** | *C++, CMake, Linux Ubuntu, GoogleTest, maroof.net* *GitHub*

- Built a **limit order book** matching engine in **C++** with **price-time priority**, processing **58 million+ orders/sec**, taking orders over my own **maroof.net** networking library to simulate a low-latency exchange.
- Benchmarked with **99.9% of orders matched in under 1 $\mu$s**, and **under 400 $\mu$s** end-to-end over UDP loopback; correctness verified by a **GoogleTest** suite. Full benchmarks in README.

**maroof.net** | *C++, CMake, Wireshark, tcpdump, Linux Ubuntu, GoogleTest* *GitHub*

- Built a low-latency **UDP/IP networking library** in **C++**, analyzed with **Wireshark** and **tcpdump**.
- Applied **data-oriented design** with techniques such as **object pools, cache-aligned lock-free SPSC queues, min-heap retransmission timers, CPU pinning, and batched syscalls**.

**NodeMonitor** | *C++, Docker, Kubernetes, GitHub Actions, PostgreSQL* *GitHub*

- Built a **network monitoring** platform in **C++** for **1,400 simulated devices** that **automates troubleshooting and remediation**, tested with a **121-case GoogleTest** suite, **Kubernetes**, and **GitHub Actions** CI/CD.

**FileSpark** | *Java, Spring Boot, AWS S3, MongoDB, Maven, Selenium, JUnit* *GitHub* | *Site*

- Launched an end-to-end file sharing app used by **100+ users**, letting people share files through links that open in a phone's native photos app instead of an unfamiliar file system.

## Technical Skills

**Languages/Methods**: C++, Python, Bash, Java, TypeScript, PostgreSQL, Agile
**Tools/Skills**: Linux, CMake, Boost, Git, GoogleTest, CMake, Wireshark, tcpdump, TCP/IP Networking, GitHub Actions, Docker, Kubernetes, AWS, JUnit, Spring Boot, MongoDB

## Education

**University of Guelph** Guelph, ON, Canada
Bachelor of Computing, Honours Computer Science | GPA: 3.7/4.0 (Dean's Honours) Expected May 2028

## Relevant Coursework

Data Structures & Algorithms, Analysis & Design of Computer Algorithms, Parallel Programming, Operating Systems, Computer Networks, Object-Oriented Programming, System Analysis and Design in Applications